TRIBHUVAN UNIVERSITY
INSTITUTE OF SCIENCE AND TECHNOLOGY
CENTRAL CAMPUS OF TECHNOLOGY (CCT), HATTISAR, DHARAN

A Internship Report On

Software Development

Submitted To:
DEPARTMENT OF IT
CENTRAL CAMPUS OF TECHNOLOGY (CCT), HATTISAR, DHARAN, SUNSARI, KOSHI, NEPAL

In partial fulfilment of the requirements for the degree of Bachelor's of Science in Computer Science and Information Technology (B.Sc. CSIT)

Submitted By:
DARSHAN DHAKAL [29592/078]

TU Registration No : 5-2-8-136-2021

Under the Supervision of:
Mr. Prakash Neupane

Falgun, 2082

INTERNSHIP CERTIFICATION

Regd. No. 149512/72/073
Information Care Pvt. Ltd.
Ward No. 10, Mahadevsthan Marg, Kathmandu

February 5, 2026

To Whom It May Concern

Subject: Internship Completion Confirmation

This letter is to inform you that **Mr. Darshan Dhakal** has successfully completed his Software Development Internship at **Information Care Pvt. Ltd.** as a **Full Stack Web Developer** from **November 2, 2025** to **February 2, 2026**.

During the internship period, he worked with and on both frontend and backend web technologies like React.js, Node.js, Express.js, PostgreSQL, Restful APIs, GraphQL. He was involved in developing user interfaces, implementing server-side functionality, developing and integrating APIs, and working with database models, relationships and queries. He performed the assigned duties sincerely and responsibly.

Throughout the internship, he showed good technical understanding, a positive learning attitude, and professionalism in the workplace. His conduct and performance were highly satisfactory during the entire internship period.

We appreciate his contribution and wish him all the best for future academic and professional endeavours.

Sincerely,

Nirmal Nyure
Project Manager
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SUPERVISOR RECOMMENDATION

This is to recommend that **DARSHAN DHAKAL [29592/078]** has carried out Internship on the position of "**Software Development**" in partial fulfilment of the requirements for the degree of Bachelor of Science (B.Sc.) degree in CSIT under my supervision in the Department of Information Technology, Central Campus of Technology, Institute of Science and Technology (IoST), Tribhuvan University (T.U.), Nepal.

To my knowledge, this work has not been submitted for any other degree. The students have fulfilled all the requirements laid down by the Institute of Science and Technology (IoST), Tribhuvan University (T.U.), Nepal for the submission of the project work for the partial fulfillment of Bachelor of Science (B.Sc.) degree in CSIT

Mr. Prakash Neupane
Supervisor
Department of IT
Central Campus of Technology
Hattisar, Dharan, Sunsari, Koshi, Nepal

CERTIFICATE OF APPROVAL

This is to certify that the internship report entitled "**Software Development** ", submitted by **Mr. DARSHAN DHAKAL [29592/078]**, a student enrolled in the Bachelor of Science (B.Sc.) degree in CSIT at the Central Campus of Technology, Dharan, has been examined and approved by the undersigned.

The report has been evaluated based on its relevance to the internship experience, presentation of facts, adherence to prescribed guidelines, and overall quality. It has been found to meet the standards set by Tribhuvan University for the partial fulfillment of the requirements for the bachelor's degree.

Mr. Prakash Neupane
Supervisor
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IOST, Tribhuvan University

Mr. Balabhadra Bhandari
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DECLARATION

This internship report, entitled "**Software Development**", is being submitted to the Department of IT, Central Campus of Technology, Institute of Science and Technology (IoST), Tribhuvan University (T.U.), Nepal, for the partial fulfillment of the requirements for the Bachelor of Science (B.Sc.) degree in CSIT. This project has been carried out by us under the supervision of Prakash Neupane, T.U., Department of IT, Central Campus of Technology, Institute of Science and Technology (IoST), Tribhuvan University (T.U.), Nepal.

We hereby declare that this work is original and has not been submitted earlier, in part or in full, to this or any other university or institution, here or elsewhere, for the award of any degree.

DARSHAN DHAKAL [29592/078]
Department of IT
Central Campus of Technology
Hattisar, Dharan, Sunsari, Koshi, Nepal

ACKNOWLEDGEMENT

First and foremost, I would like to express my sincere gratitude to **Information Care Pvt. Ltd.** for providing me with the opportunity to undertake my internship as a **Full Stack Web Developer**. This internship provided valuable industry exposure and practical experience, allowing me to strengthen my technical knowledge and professional skills in a real-world working environment.

I am deeply grateful to my mentor, **Mr. Aashish Pokhrel**, for his continuous guidance, technical support, and encouragement throughout the internship period. His mentorship played a vital role in enhancing my understanding of web development and software development practices.

I would also like to extend my sincere thanks to my supervisor, **Mr. Prakash Neupane**, for his support, constructive feedback, and supervision during the internship. His guidance greatly contributed to the successful completion of my assigned tasks and projects.

Furthermore, I would like to thank the faculty members of the **Department of Computer Science and Information Technology**, **Institute of Science and Technology (IoST)**, **Tribhuvan University**, for their academic guidance and support throughout my studies.

I am thankful to my colleagues, friends, and fellow students for their encouragement and support during this internship. I am also sincerely grateful to my family for their unwavering support and motivation throughout this journey.

Finally, I would like to thank everyone who directly or indirectly contributed to making this internship a valuable and enriching learning experience.

Thank you.

DARSHAN DHAKAL [29592/078]

ABSTRACT

This report presents the work carried out during the internship as a **Full Stack Web Developer** at **Information Care Pvt. Ltd.**. The internship was conducted with the objective of gaining practical exposure to full-stack application development, system architecture design, and modern web technologies in a professional environment.

During the internship period, various frontend and backend technologies were studied and implemented to design, develop, and maintain scalable web applications. On the frontend, responsive and user-centric interfaces were developed using modern JavaScript frameworks and component-based architecture principles. On the backend, RESTful APIs were designed and implemented with proper routing, authentication, and database integration to ensure secure and efficient data management.

Major activities included designing database schemas, implementing CRUD operations, integrating third-party APIs, managing authentication and authorization mechanisms, and optimizing application performance. The internship also involved version control using Git, collaborative development workflows, debugging, and deployment of applications to cloud environments.

In addition, tasks related to testing, performance optimization, and security enhancement were performed to ensure application reliability and robustness. Emphasis was placed on writing clean, maintainable code following industry best practices and software development lifecycle methodologies.

Overall, the internship provided practical knowledge of full-stack development, system integration, and real-world software engineering practices, helping to bridge the gap between academic learning and industry-level application development.

Keywords: Full Stack Development, Web Applications, Frontend, Backend, REST API, Database, Authentication, Deployment.

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LIST OF ABBREVIATIONS

AWS Amazon Web Services.

CI/CD Continuous Integration and Continuous Delivery.

DKIM DomainKeys Identified Mail.

EC2 Elastic Compute Cloud.

GUMP Genese Unified Mailing Platform.

IaC Infrastructure as Code.

IAM Identity and Access Management.

RDS Relational Database Service.

S3 Simple Storage Service.

SAM Serverless Application Model.

SMTP Simple Mail Transfer Protocol.

SSM Systems Manager.

WAF Web Application Firewall.

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CHAPTER 1
INTRODUCTION

1.1. Introduction

The rapid growth of web technologies has significantly transformed how modern software applications are designed and developed. Today's applications are expected to be interactive, scalable, secure, and responsive across different devices. This shift has increased the demand for full stack developers who can work on both the frontend and backend of web systems and understand how all components integrate into a complete application.

DevOps is not merely a toolset but a philosophy that emphasizes automation, monitoring, version control, infrastructure as code (IaC), containerization, and cloud-based deployment. Combined with cloud computing platforms such as Amazon Web Services (AWS), DevOps enables organizations to build scalable, resilient, and highly available systems with faster release cycles and improved reliability. [1]

During my internship at Genese Solution Pvt. Ltd., Lalitpur, Nepal, I worked as a Cloud and DevOps Engineering Intern, where I gained hands-on experience in implementing DevOps practices within real-world production environments. The organization specializes in cloud-based infrastructure solutions, automation, and scalable system design. Throughout the internship period, I was actively involved in designing and implementing CI/CD pipelines, managing AWS resources, automating deployment processes using AWS SAM (Serverless Application Model), configuring IAM roles and policies, and working with services such as AWS Lambda, CodeBuild, CodePipeline, S3, CloudFormation, EventBridge, and CloudWatch.

One of the most valuable aspects of this internship was understanding how different layers of an application communicate with each other. I learned how to structure backend projects properly, organize routes and controllers, manage middleware, design normalized database schemas, and ensure secure data transmission between client and server. Working in a collaborative environment also strengthened my skills in Git-based version control, debugging, and writing clean, maintainable code.

This internship helped me connect theoretical concepts learned in academic courses with real-world software development practices. It improved my understanding of software development lifecycle (SDLC), system design principles, REST architecture, and performance optimization in full stack applications.

1.2. Problem Statement

While developing modern web applications, several practical challenges can arise. Many traditional systems suffer from tightly coupled architectures, poor database design, and weak security mechanisms. Without a structured development approach, applications may become difficult to maintain, scale, or extend.

During full stack development, common technical challenges include:

- Designing a clear separation between frontend and backend logic
- Managing database relationships and optimizing queries
- Implementing secure authentication and authorization mechanisms
- Handling asynchronous operations in backend services
- Managing application state effectively in frontend applications
- Ensuring proper validation and error handling

Organizations aiming to build cloud-based applications often struggle with:

- Automating deployments efficiently
- Managing infrastructure securely
- Maintaining environment consistency
- Implementing continuous integration and continuous deployment (CI/CD)
- Managing multiple microservices or serverless functions efficiently
- Ensuring security and least-privilege access using IAM policies

Specifically, in serverless monorepo architectures, challenges arise in:

- Handling shared dependencies (common layers)
- Triggering selective deployments
- Avoiding unnecessary rebuilds
- Maintaining version control for multiple Lambda functions
- Ensuring proper role-based access control

- Managing S3 artifacts and CloudFormation packaging

Therefore, there was a need to design and implement a structured DevOps workflow that would:

- Automate build, package, and deployment processes
- Enable scalable serverless architecture
- Reduce manual intervention
- Improve deployment reliability
- Enforce security best practices
- Optimize cloud resource usage

1.3. Objectives

To understand and implement DevOps practices using cloud technologies in a production-level environment.

1.4. Scope and Limitation

Scope The scope of this internship primarily covered:

- Frontend development using React.js with reusable component design
- Backend API development using Node.js and Express.js
- PostgreSQL database design and query implementation
- Implementation of CRUD operations
- Authentication and authorization mechanisms
- Git-based version control and collaborative development
- Basic deployment and application testing

The experience provided exposure to real development environments where applications are built based on actual business requirements. It also improved my understanding of how to write production-level code and maintain coding standards within a team.

The internship is directly related to academic subjects such as Software Engineering, Database Management Systems, Web Technology, and Object-Oriented Programming.

Limitation Despite the extensive exposure, certain limitations were present:

- The internship duration was limited.
- Access to highly sensitive production data was restricted.
- Exposure to large-scale enterprise architectures was limited.
- The primary focus remained on the PERN stack rather than multiple technology stacks

Some advanced topics such as large-scale distributed systems, enterprise-level DevOps automation, and high-availability system design were beyond the scope of this internship.

1.5. Report Organization

This report is organized into several chapters to clearly present the internship experience and technical contributions:

- **Chapter 1: Introduction**
  Provides background, objectives, problem statement, scope, and structure of the report.
- **Chapter 2: Organization Overview**
  Describes Genese Solution Pvt. Ltd., its services, working environment, and organizational structure.
- **Chapter 3: Literature Review / Background Study**
  Covers theoretical concepts related to DevOps, CI/CD, Cloud Computing, AWS services, Infrastructure as Code, and Serverless Architecture.
- **Chapter 4: Methodology and Tools Used**
  Explains the technologies, tools, and workflow implemented during the internship.
- **Chapter 5: System Design and Implementation**
  Describes the architecture, CI/CD pipelines, serverless deployment structure, IAM configuration, and automation strategies.
- **Chapter 6: Results and Discussion**
  Analyzes implementation outcomes, performance improvements, challenges faced, and solutions applied.
- **Chapter 7: Conclusion and Future Recommendations**
  Summarizes key learning outcomes and suggests possible improvements and future enhancements.

- **Appendices**
  Includes screenshots, configuration files, code snippets, pipeline diagrams, and additional documentation.

CHAPTER 2
ORGANIZATION DETAILS AND LITERATURE REVIEW

2.1. Introduction to Organization

Genese Solution Pvt. Ltd. is a multinational technology consulting and cloud services company with a strong presence in Nepal and several other countries across Europe and Asia. It operates as part of the larger Genese Solution group headquartered in the United Kingdom, offering a wide range of digital transformation solutions including cloud computing consultation, software development, cybersecurity, infrastructure automation, and DevOps services. The company specializes in helping organizations optimize their IT operations by leveraging modern cloud technologies such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform to build scalable, secure, and cost-efficient systems. With a focus on innovation and customer-centric solutions, Genese Solution works with enterprises, startups, and public sector clients to design and deploy end-to-end digital solutions tailored to their unique business needs. [2]

In Nepal, Genese Solution has established itself as a key player in the IT and cloud consulting landscape, operating out of Jhamsikhel, Lalitpur and contributing significantly to the adoption of modern cloud and DevOps practices within the region. The company's team comprises highly skilled and internationally certified professionals who bring deep technical expertise to projects involving cloud migration, DevOps automation, continuous integration and delivery (CI/CD), and infrastructure optimization. Through its services, Genese Solution helps organizations enhance operational efficiency, reduce manual intervention, and accelerate software delivery cycles. Additionally, the company engages in initiatives aimed at bridging the skill gap between academia and industry by supporting cloud education and workforce development, further strengthening Nepal's technology ecosystem. [2]

2.2. Organizational Hierarchy

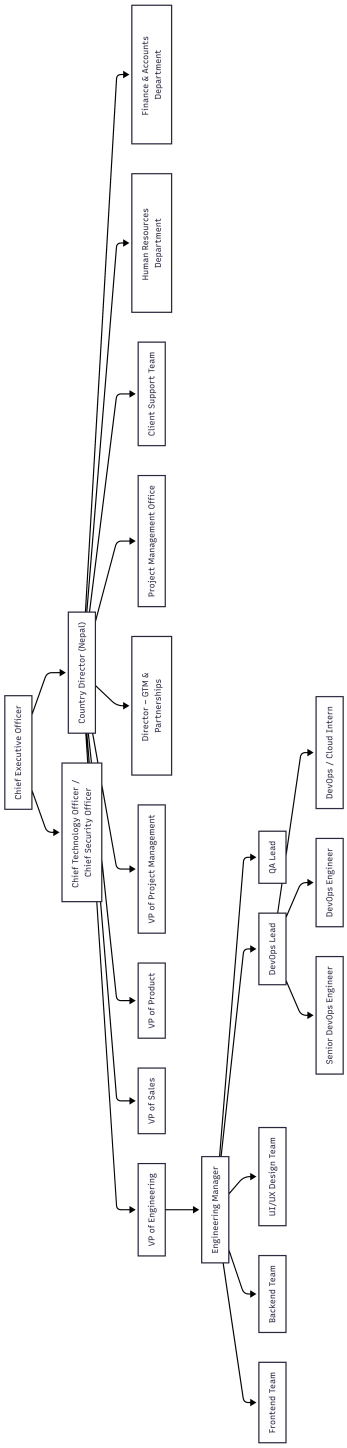

Figure 2.1: Organizational structure of Genese Solution Pvt. Ltd.

2.3. Working Domains of Organization

Genese Solution operates across multiple technology and consulting domains with a focus on enabling digital transformation and cloud adoption for businesses of all sizes. The primary working domains of the organization include:

- **Cloud Consulting & Cloud Services:** Providing strategy, planning, migration, managed support, and optimization of cloud infrastructure, helping organizations transition to scalable and cost-efficient cloud platforms.
- **DevOps and Cloud Automation:** Implementing modern DevOps practices including continuous integration and continuous deployment (CI/CD), infrastructure as code (IaC), automated deployment pipelines, monitoring, and operational automation.
- **Software and Application Development:** Delivering web and mobile application solutions, backend systems, API development, and customized software to support business digitalization.
- **Cybersecurity & Compliance Solutions:** Offering security assessments, compliance audits, cyber risk mitigation strategies, and secure architecture services to protect critical systems.
- **Productivity & Collaboration Tools:** Deployment and support of communication and productivity systems for enterprises, such as unified collaboration suites and monitoring tools.
- **Digital Marketing & SEO Services:** Helping businesses improve online presence and engagement through search engine optimization and digital marketing strategies.
- **Cloud Education & Skill Development (Genese Cloud Academy):** Training programs and mentorship aimed at bridging the gap between academic knowledge and industry requirements by equipping students and professionals with cloud computing and DevOps skills.
- **Domain and Web Hosting Services:** Providing domain registration, hosting solutions, and managed server environments tailored to business needs.
- **Monitoring and Performance Optimization:** Tools and services dedicated to monitoring infrastructure performance, identifying anomalies, and optimizing cloud resource usage.

The organization's expertise spans a wide spectrum of digital technology domains, enabling it to support clients across different industries by delivering secure, scalable, and future-ready technology solutions.

2.4. Description of Intern Department

During my internship at Genese Solution Pvt. Ltd., Nepal, I was placed within the Cloud & DevOps Engineering Department, a core technical division responsible for implementing automation, cloud

infrastructure solutions, and modern software delivery processes. This department plays a pivotal role in transforming how the company and its clients manage software deployment, infrastructure provisioning, and operational scalability. It focuses on designing, deploying, and maintaining cloud-native systems primarily on Amazon Web Services (AWS), as well as integrating continuous integration and continuous delivery (CI/CD) pipelines to streamline the development lifecycle. The team works collaboratively with cross-functional stakeholders to ensure production-grade automation, robust security practices, and highly available cloud architectures, all of which contribute directly to modern digital transformation goals.

2.5. Literature Review

DevOps is a modern software engineering paradigm that emerged to address the inefficiencies and disconnects between traditional software development and IT operations. At its core, DevOps emphasizes collaboration, shared responsibility, and continuous processes that span from initial code development to production deployment and operational monitoring. Studies define DevOps as a cultural and technical approach that integrates development and operations teams to improve software delivery performance while reducing errors and lead time for changes. DevOps adoption involves organizational transformation as much as technical change, requiring clear practices, guidelines, and close alignment of cross-functional teams to realize its full potential. Research has shown that adopting DevOps practices can significantly streamline processes, improve communication, and enhance overall software delivery quality and velocity. [3]

A foundational element of DevOps is the implementation of continuous practices such as Continuous Integration (CI) and Continuous Delivery/Deployment (CD). CI involves the frequent integration of code changes into a shared repository, triggering automated builds and tests to detect issues early in the software lifecycle. CD extends CI by automating the release and deployment of validated changes to production environments with minimal manual intervention, improving delivery speed and system reliability. Research in this area highlights that CI/CD pipelines reduce human errors, accelerate feedback loops, and support faster feature deliveries, which are critical for competitive advantage in today's software-driven markets.[1]

The role of cloud computing has become particularly significant in enabling DevOps practices at scale. The elasticity, automation, and on-demand infrastructure of cloud platforms provide ideal underpinnings for DevOps workflows, allowing teams to dynamically provision resources, scale testing environments, and manage releases across multiple environments. Cloud-enabled DevOps has been linked with higher deployment frequency, improved scalability, and better resource utilization. Nevertheless, literature also notes that the integration of cloud infrastructure within DevOps introduces new challenges related to security, cost management, and the handling of legacy systems, indicating that organizations must adopt structured strategies to address these concerns.[4]

Another critical practice within DevOps is Infrastructure as Code (IaC), which describes the automa-

tion of infrastructure provisioning and configuration through code rather than manual processes. IaC improves consistency, repeatability, and traceability of infrastructure changes, helping teams manage complex environments reliably.This aligns with modern DevOps objectives of automating as much of the software delivery lifecycle as possible, further enabling rapid, scalable deployments. [5]

CHAPTER 3
INTERNSHIP ACTIVITIES

3.1. Roles and Responsibilities

During my internship at **Genese Solution Pvt. Ltd.**, I performed the following roles and responsibilities in the Cloud & DevOps Engineering Department:

- Assisted in designing and implementing **CI/CD pipelines** using AWS CodePipeline and AWS CodeBuild.
- Automated build, packaging, and deployment processes for serverless applications.
- Worked with **AWS SAM (Serverless Application Model)** for building and deploying Lambda-based applications.
- Developed and modified **AWS CloudFormation templates** to provision infrastructure using Infrastructure as Code (IaC).
- Managed deployment artifacts in **Amazon S3**, including packaging and version control of application builds.
- Configured and maintained **AWS IAM roles and policies** following the principle of least privilege.
- Assisted in creating and managing **serverless architectures using AWS Lambda** in a monorepo structure.
- Handled shared Lambda layers, versioning, and dependency management.
- Monitored application logs and system performance using **Amazon CloudWatch**.
- Identified and troubleshot build failures, deployment errors, and configuration issues.
- Supported multi-environment deployments (Development, Staging, Production).
- Participated in reviewing infrastructure changes and deployment workflows.
- Collaborated with development teams to improve automation and deployment efficiency.
- Maintained technical documentation related to CI/CD pipelines, IAM configuration, and deployment procedures.
- Followed Agile methodologies and participated in team meetings and progress discussions.
- Ensured adherence to cloud security best practices and operational standards.

3.2. Weekly log

Table 3.1: Weekly log of activities performed during the internship.

Week	Activity Performed
Week 1	Project onboarding and AWS architecture review of GUMP Now (Lambda, RDS, EC2, CloudWatch, CI/CD).
Week 2	Research on CloudWatch Synthetics canaries for availability, latency, and user-journey monitoring.
Week 3	Design of CloudWatch on-call support dashboards for staging and production environments.
Week 4	Implementation of CloudWatch dashboards and monitoring metrics for operational visibility.
Week 5	Implementation of IAM-based authentication for RDS in a staging environment.
Week 6	Research and execution of AWS SSM Patch Manager to patch all EC2 instances in staging.
Week 7	Research on AWS WAF capabilities including managed rules and web attack mitigation strategies.
Week 8	Design of automated CI/CD pipeline for Lambda monorepo using AWS SAM and custom trigger Lambda.
Week 9	Implementation of automated Lambda monorepo pipeline with CodeBuild, SAM build, package, and deploy.
Week 10	SMTP server proof of concept using Node.js `smtp-server` package for mail flow analysis.
Week 11	Functional and usability testing of the official GUMP Now website.
Week 12	Production improvements: Outlook invitation email fix and DKIM CSV download feature deployment to staging and production.

3.3. Description of the Project Involved During Internship

The **Genese Unified Mailing Platform (GUMP)** is an enterprise-grade cloud-based mailing solution developed by **Genese Solution Pvt. Ltd.** to streamline and centralize email communication processes for organizations. The platform is designed to handle high-volume email delivery requirements, including transactional emails, notification systems, marketing campaigns, and automated communication workflows. GUMP provides a unified interface for managing email templates, user segmentation, scheduling, tracking, and reporting, ensuring reliable and efficient communication between organizations and their customers. [6]

During my internship, I worked on multiple technical projects related to the **GUMP Now** platform, primarily focusing on cloud infrastructure, security enhancement, monitoring, automation,

and operational support. One of the key contributions involved designing and implementing cloud monitoring and observability solutions using **AWS CloudWatch**, including dashboards and synthetic monitoring for improved availability tracking and proactive issue detection in staging and production environments.

I also contributed to cloud security improvements by implementing **IAM-based authentication for Amazon RDS** and strengthening system security through patch management of EC2 instances using **AWS SSM Patch Manager**. In addition, I designed and implemented an automated **CI/CD pipeline** for a Lambda monorepo using **AWS SAM** and a custom trigger Lambda function, enabling efficient and selective serverless deployments.

Furthermore, I supported email infrastructure and application-level enhancements, including an SMTP server proof of concept, resolving Outlook email rendering issues, implementing a DKIM CSV download feature, conducting website testing, and supporting both staging and production environments. These activities collectively aimed to enhance the reliability, security, and operational efficiency of the GUMP platform.

3.4. Activities Performed

- **Cloud Monitoring and Observability:** Researched and configured AWS CloudWatch features, including custom dashboards and performance metrics for staging and production environments. Evaluated CloudWatch Synthetics canaries to monitor application availability, response time, and end-to-end user journeys, supporting proactive incident detection and on-call operations.
- **RDS IAM Authentication Implementation:** Implemented IAM-based authentication for Amazon RDS in the Gumpnow staging environment. Configured IAM roles and policies, updated database connection logic, and validated secure database access without static credentials.
- **On-Call Support Dashboards:** Designed and deployed CloudWatch dashboards aggregating key service metrics such as health status, error rates, and alarms. Validated dashboards in both staging and production environments.
- **Website and Application Testing:** Conducted functional and usability testing of the official GUMP Now website to ensure correct behavior, identify defects, and validate fixes before and after deployments.
- **SMTP Server Proof of Concept:** Developed a proof-of-concept SMTP server using the Node.js `smtp-server` package to analyze email handling, message flow, and integration mechanisms within the email infrastructure.
- **AWS WAF Research:** Researched AWS Web Application Firewall (WAF) capabilities, including managed rule groups and custom rules, to evaluate protection mechanisms against

SQL injection, cross-site scripting (XSS), and other web vulnerabilities.

- **Patch Management Using AWS SSM:** Patched all EC2 instances in the Gumpnow staging environment using AWS Systems Manager Patch Manager. Verified compliance, monitored execution, and ensured system stability after patching.
- **Email Rendering and Deliverability Fixes:** Investigated and resolved an issue where invitation email buttons were not displayed correctly in Microsoft Outlook. Improved HTML/CSS compatibility and validated rendering across email clients.
- **DKIM Feature Enhancement:** Implemented a feature in the `gumpnow-frontend-app` to enable downloading DKIM records in CSV format. Deployed and verified the feature in staging and production environments.
- **CI/CD Automation for Lambda Monorepo:** Designed and implemented an automated CI/CD pipeline for a Lambda monorepo using AWS SAM. Developed a custom trigger Lambda function to detect code changes and selectively trigger deployments, improving efficiency and reducing unnecessary builds.

CHAPTER 4
CONCLUSION AND LEARNING OUTCOMES

4.1. Conclusion

The internship at Genese Solution Pvt. Ltd. provided valuable practical exposure to real-world cloud infrastructure management, DevOps practices, and enterprise application support. Working on the Genese Unified Mailing Platform (GUMP Now) allowed me to understand how modern cloud-native systems are designed, deployed, secured, and maintained in production environments.

Throughout the internship period, I contributed to monitoring and observability improvements, cloud security enhancements, CI/CD automation, and operational support tasks. The implementation of CloudWatch dashboards, IAM-based RDS authentication, EC2 patch management using AWS Systems Manager, and Lambda monorepo automation significantly improved system reliability, security posture, and deployment efficiency. Additionally, involvement in email infrastructure improvements, DKIM feature enhancements, and frontend application support provided insight into how infrastructure and application layers interact in a production system.

This internship successfully bridged the gap between academic theory and industry practice. It strengthened my understanding of distributed systems, cloud computing concepts, security best practices, and automation workflows. The experience enhanced my ability to troubleshoot real-world problems, design scalable solutions, and contribute effectively within a collaborative engineering environment. Overall, the internship was instrumental in shaping my technical competence and professional readiness as a Cloud and DevOps Engineer.

4.2. Learning Outcome

During the internship, I achieved the following key learning outcomes:

- Developed practical understanding of DevOps lifecycle and CI/CD pipelines in real production environments.
- Gained hands-on experience with Amazon Web Services (AWS) including Lambda, EC2, RDS, S3, CloudWatch, IAM, WAF, and Systems Manager.
- Learned to implement Infrastructure as Code (IaC) using AWS SAM and CloudFormation.
- Understood the importance of monitoring and observability, including dashboard design, metrics tracking, and synthetic monitoring.

- Strengthened knowledge of cloud security best practices, including IAM role configuration, RDS IAM authentication, and web application firewall strategies.
- Acquired experience in automated patch management and maintaining system compliance.
- Improved skills in troubleshooting deployment failures and operational issues in staging and production environments.
- Learned to design and implement serverless architectures and Lambda-based monorepo pipelines.
- Enhanced understanding of email infrastructure, DKIM authentication, SMTP workflows, and email client compatibility.
- Developed professional skills such as team collaboration, technical documentation, agile work practices, and communication.
- Gained confidence in working with enterprise-level systems and production-grade cloud environments.

  Overall, the internship contributed significantly to both my technical expertise and professional development, preparing me for future roles in Cloud Engineering and DevOps domains.

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APPENDIX

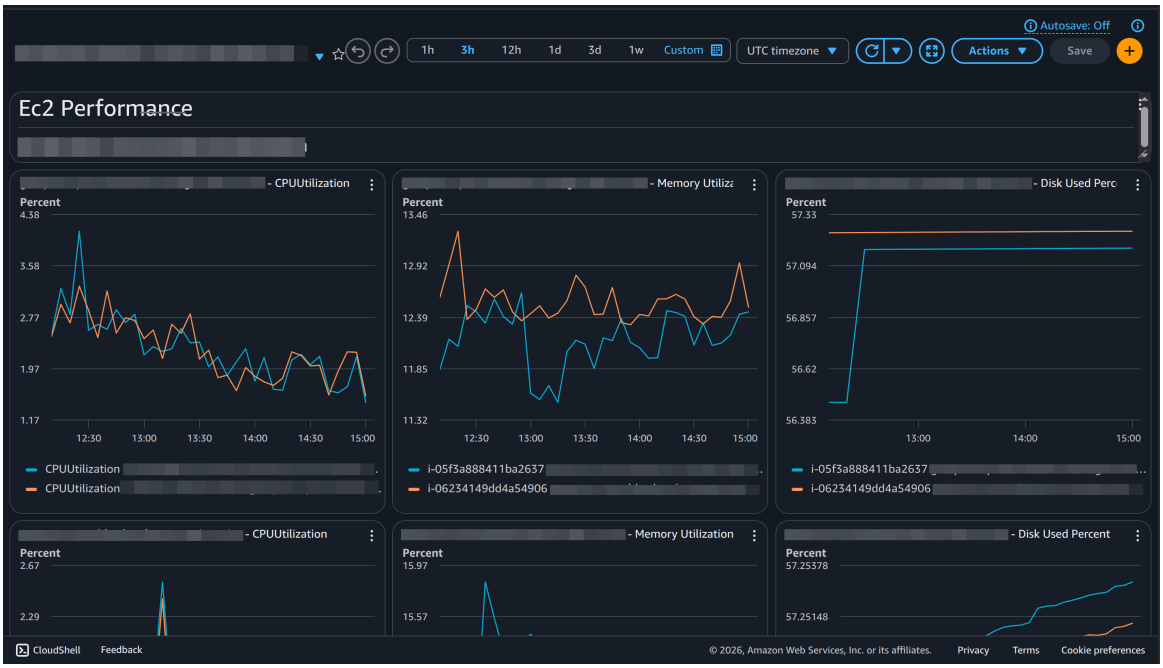

Figure 4.1: CloudWatch on-call support dashboards

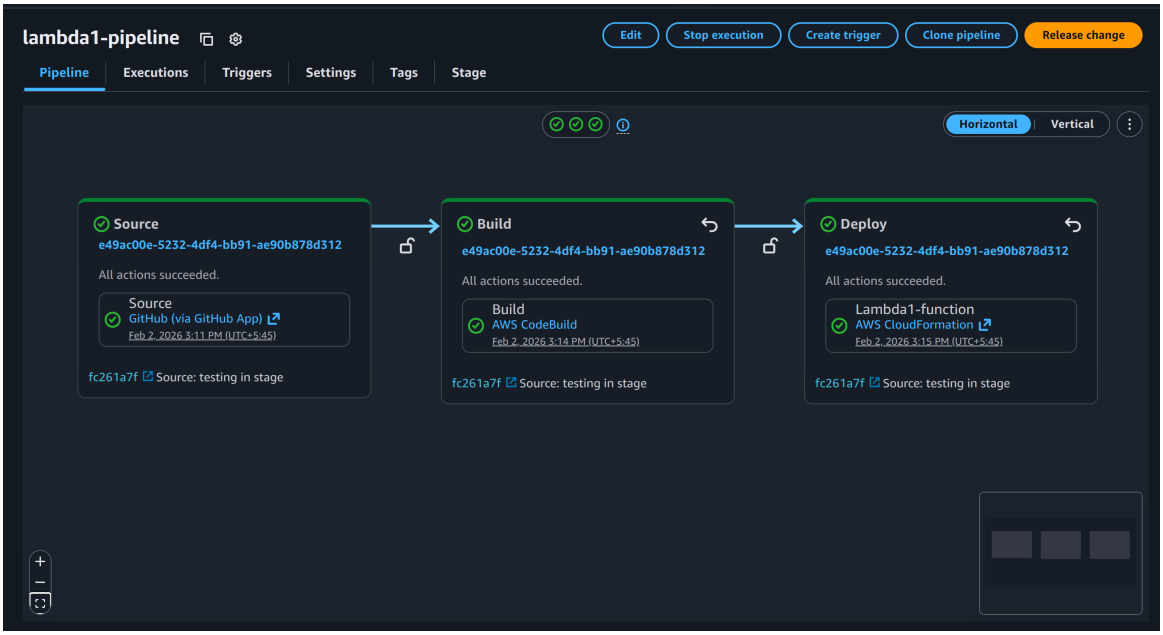

Figure 4.2: Automated CI/CD pipeline for Lambda monorepo

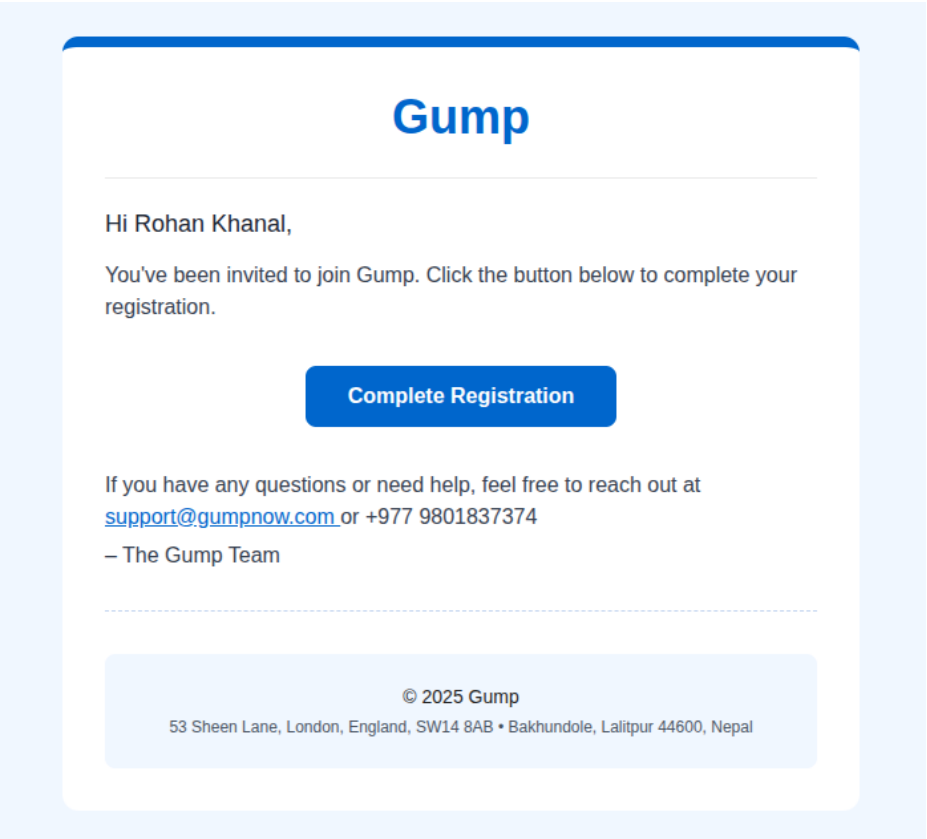

Figure 4.3: Outlook invitation email fix

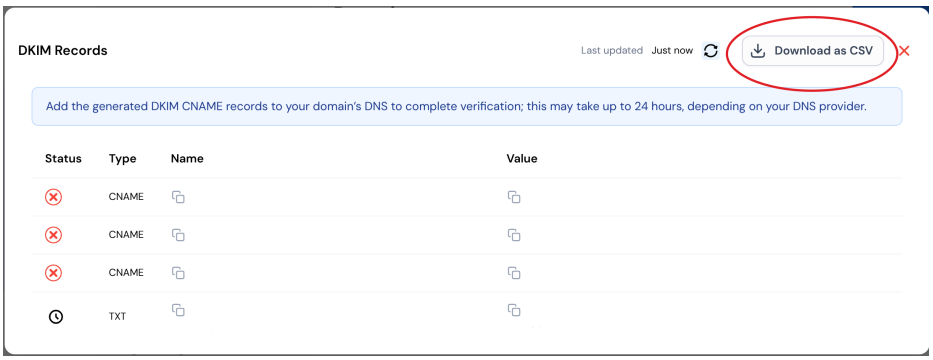

Figure 4.4: DKIM CSV download button

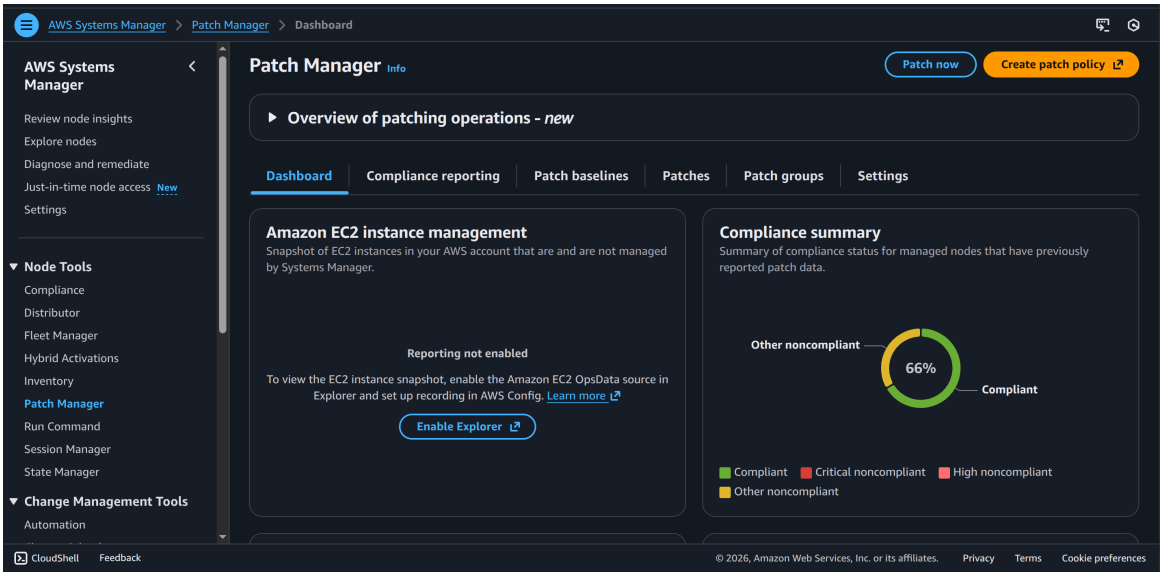

Figure 4.5: AWS SSM Patch Manager

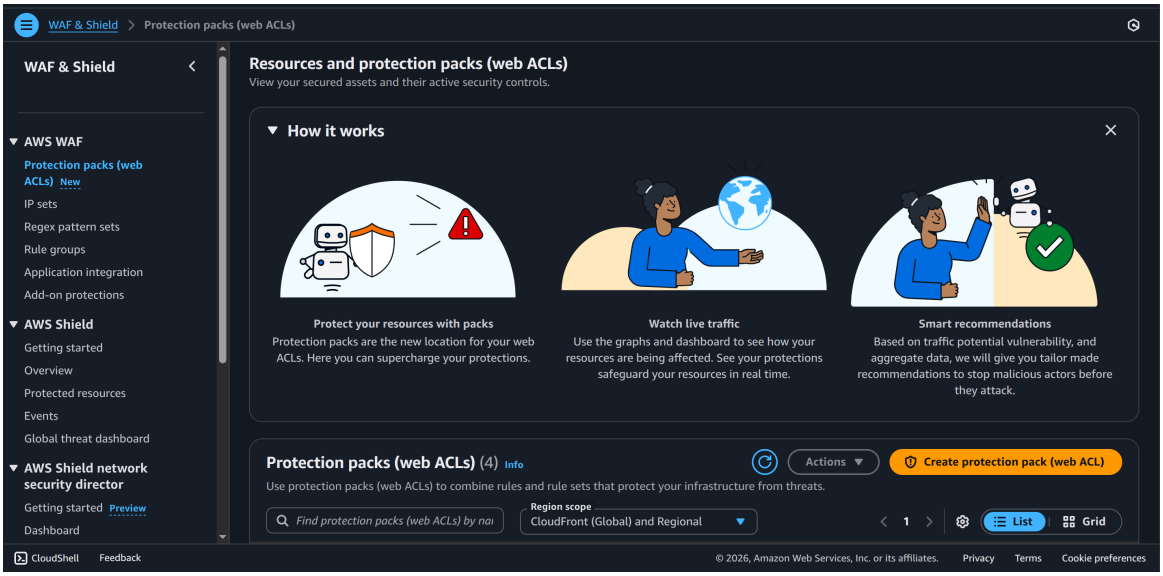

Figure 4.6: AWS WAF

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Weekly Internship Logbook

Week No: 1
Name of Student: Darshan Dhakal
Internship Site/ Organization: Information Care Pvt. Ltd
Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-07-24	9	9	Internship orientation and introduction to GUMP Now project.
2	2082-07-25	9	18	Reviewed AWS architecture (Lambda, EC2, RDS, CloudWatch).
3	2082-07-26	9	27	Studied staging and production environments.
4	2082-07-27	9	36	Analyzed CI/CD pipelines and deployment flow.
5	2082-07-28	9	45	Documentation of system architecture and learning notes.
6	2082-07-29	Holiday	45	Weekend
7	2082-07-30	Holiday	45	Weekend

Current Week Summary: Completed onboarding and gained understanding of AWS-based architecture.

Next Week Plan: Research CloudWatch Synthetics.

Total Hours Worked: 45

Student's Signature: **Date:** 2082-10-16

Mentor's Signature: **Date:** 2082-10-16

Supervisor's Signature: _____ **Date:** _____

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Weekly Internship Logbook

Week No: 2

Name of Student: Darshan Dhakal

Internship Site/ Organization: Information Care Pvt. Ltd

Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-08-01	9	54	Research on CloudWatch Synthetics features.
2	2082-08-02	9	63	Studied availability monitoring using canaries.
3	2082-08-03	9	72	Tested latency monitoring setup.
4	2082-08-04	9	81	Created sample user-journey canary.
5	2082-08-05	9	90	Evaluated alarm integration and reporting.
6	2082-08-06	Holiday	90	Weekend
7	2082-08-07	Holiday	90	Weekend

Current Week Summary: Evaluated CloudWatch Synthetics suitability.

Next Week Plan: Design monitoring dashboards.

Total Hours Worked: 45

Student's Signature: **Date:** 2082-10-16

Mentor's Signature: **Date:** 2082-10-16

Supervisor's Signature: _____ **Date:** _____

Central Campus of Technology

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Weekly Internship Logbook

Week No: 3

Name of Student: Darshan Dhakal

Internship Site/ Organization: Information Care Pvt. Ltd

Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-08-08	9	99	Identified key operational metrics for staging and production.
2	2082-08-09	9	108	Designed CloudWatch dashboard layout and widget structure.
3	2082-08-10	9	117	Planned Lambda/EC2/RDS metrics grouping and visualization.
4	2082-08-11	9	126	Reviewed alarm requirements and on-call visibility needs.
5	2082-08-12	9	135	Documented dashboard design and implementation plan.
6	2082-08-13	Holiday	135	Weekend
7	2082-08-14	Holiday	135	Weekend

Current Week Summary: Designed CloudWatch dashboards by structuring visualization for improved observability.

Next Week Plan: Implement dashboards and configure monitoring alarms.

Total Hours Worked: 45

Student's Signature: **Date:** 2082-10-16

Mentor's Signature: **Date:** 2082-10-16

Supervisor's Signature: _____ **Date:** _____

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Weekly Internship Logbook

Week No: 4

Name of Student: Darshan Dhakal

Internship Site/ Organization: Information Care Pvt. Ltd

Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-08-15	9	144	Implemented CloudWatch dashboard for staging environment.
2	2082-08-16	9	153	Implemented CloudWatch dashboard for production environment.
3	2082-08-17	9	162	Configured CloudWatch alarms for key metrics.
4	2082-08-18	9	171	Tested alert notifications and verified alarm triggers.
5	2082-08-19	9	180	Fine-tuned dashboard widgets and documented monitoring setup.
6	2082-08-20	Holiday	180	Weekend
7	2082-08-21	Holiday	180	Weekend

Current Week Summary: Successfully implemented dashboards and alarms to enhance monitoring.

Next Week Plan: Strengthen database security using IAM authentication.

Total Hours Worked: 45

Student's Signature: **Date:** 2082-10-16

Mentor's Signature: **Date:** 2082-10-16

Supervisor's Signature: _____ **Date:** _____

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Weekly Internship Logbook

Week No: 5

Name of Student: Darshan Dhakal

Internship Site/ Organization: Information Care Pvt. Ltd

Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-08-22	9	189	Researched RDS IAM authentication workflow and requirements.
2	2082-08-23	9	198	Configured IAM roles and policies for DB authentication.
3	2082-08-24	9	207	Enabled IAM database authentication in staging.
4	2082-08-25	9	216	Tested secure database connection using IAM tokens.
5	2082-08-26	9	225	Removed static DB credentials and documented security improvements.
6	2082-08-27	Holiday	225	Weekend
7	2082-08-28	Holiday	225	Weekend

Current Week Summary: Implemented IAM-based RDS authentication to enhance database security.

Next Week Plan: Research and implement patch management using AWS SSM.

Total Hours Worked: 45

Student's Signature: **Date:** 2082-10-16

Mentor's Signature: **Date:** 2082-10-16

Supervisor's Signature: _____ **Date:** _____

Central Campus of Technology

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Weekly Internship Logbook

Week No: 6

Name of Student: Darshan Dhakal

Internship Site/ Organization: Information Care Pvt. Ltd

Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-08-29	9	234	Studied AWS SSM Patch Manager concepts and workflow.
2	2082-09-01	9	243	Configured patch baseline and maintenance strategy.
3	2082-09-02	9	252	Performed EC2 patch scan and reviewed missing patches.
4	2082-09-03	9	261	Patched EC2 instances in staging environment.
5	2082-09-04	9	270	Verified compliance status and generated reports.
6	2082-09-05	Holiday	270	Weekend
7	2082-09-06	Holiday	270	Weekend

Current Week Summary: Successfully patched EC2 instances using AWS SSM Patch Manager.

Next Week Plan: Research AWS WAF for web application security.

Total Hours Worked: 45

Student's Signature: **Date:** 2082-10-16

Mentor's Signature: **Date:** 2082-10-16

Supervisor's Signature: _____ **Date:** _____

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Weekly Internship Logbook

Week No: 7

Name of Student: Darshan Dhakal

Internship Site/ Organization: Information Care Pvt. Ltd

Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-09-07	9	279	Researched AWS WAF capabilities and deployment options.
2	2082-09-08	9	288	Studied AWS Managed Rule Groups and best practices.
3	2082-09-09	9	297	Evaluated SQL injection protection approach.
4	2082-09-10	-	297	Christmas Leave - No work done.
5	2082-09-11	9	306	Reviewed rate limiting and documented recommendations.
6	2082-09-12	Holiday	306	Weekend
7	2082-09-13	Holiday	306	Weekend

Current Week Summary: Researched AWS WAF features for securing GUMP Now against web threats.

Next Week Plan: Design automated CI/CD pipeline for Lambda monorepo.

Total Hours Worked: 36

Student's Signature: **Date:** 2082-10-16

Mentor's Signature: **Date:** 2082-10-16

Supervisor's Signature: _____ **Date:** _____

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Weekly Internship Logbook

Week No: 8

Name of Student: Darshan Dhakal

Internship Site/ Organization: Information Care Pvt. Ltd

Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-09-14	9	315	Designed automated Lambda monorepo CI/CD architecture.
2	2082-09-15	9	324	Created custom trigger Lambda for selective deployment.
3	2082-09-16	9	333	Designed SAM build workflow and repository structure.
4	2082-09-17	-	333	New Year Leave - No work done.
5	2082-09-18	9	342	Planned environment-based deployment strategy.
6	2082-09-19	Holiday	342	Weekend
7	2082-09-20	Holiday	342	Weekend

Current Week Summary: Designed CI/CD pipeline architecture for Lambda monorepo automation.

Next Week Plan: Implement and test pipeline deployment workflow.

Total Hours Worked: 36

Student's Signature: **Date:** 2082-10-16

Mentor's Signature: **Date:** 2082-10-16

Supervisor's Signature: _____ **Date:** _____

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Weekly Internship Logbook

Week No: 9

Name of Student: Darshan Dhakal

Internship Site/ Organization: Information Care Pvt. Ltd

Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-09-21	9	351	Implemented CodeBuild pipeline stages.
2	2082-09-22	9	360	Integrated SAM build and packaging process.
3	2082-09-23	9	369	Integrated SAM deploy process.
4	2082-09-24	9	378	Debugged deployment configuration issues.
5	2082-09-25	9	387	Validated staging and production deployments.
6	2082-09-26	Holiday	387	Weekend
7	2082-09-27	Holiday	387	Weekend

Current Week Summary: Successfully implemented and tested automated CI/CD pipeline.

Next Week Plan: Explore SMTP server and email handling mechanisms.

Total Hours Worked: 45

Student's Signature: **Date:** 2082-10-16

Mentor's Signature: **Date:** 2082-10-16

Supervisor's Signature: _____ **Date:** _____

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Weekly Internship Logbook

Week No: 10

Name of Student: Darshan Dhakal

Internship Site/ Organization: Information Care Pvt. Ltd

Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-09-28	9	396	Researched SMTP protocol and email flow basics.
2	2082-09-29	9	405	Built SMTP server proof of concept using Node.js.
3	2082-09-30	9	414	Tested mail flow and analyzed server behavior.
4	2082-10-01	9	423	Analyzed email headers and logs.
5	2082-10-02	9	432	Documented findings and implementation insights.
6	2082-10-03	Holiday	432	Weekend
7	2082-10-04	Holiday	432	Weekend

Current Week Summary: Conducted SMTP proof of concept and analyzed email handling workflow.

Next Week Plan: Perform functional and usability testing of website.

Total Hours Worked: 45

Student's Signature: **Date:** 2082-10-16

Mentor's Signature: **Date:** 2082-10-16

Supervisor's Signature: _____ **Date:** _____

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Weekly Internship Logbook

Week No: 11

Name of Student: Darshan Dhakal

Internship Site/ Organization: Information Care Pvt. Ltd

Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-10-05	9	441	Performed functional testing of GUMP Now website.
2	2082-10-06	9	450	Conducted usability evaluation.
3	2082-10-07	9	459	Identified UI bugs and issues.
4	2082-10-08	9	468	Verified fixes and improvements.
5	2082-10-09	9	477	Prepared QA documentation.
6	2082-10-10	Holiday	477	Weekend
7	2082-10-11	Holiday	477	Weekend

Current Week Summary: Performed functional and usability testing, identified UI issues, and validated fixes.

Next Week Plan: Implement final production improvements and complete internship documentation.

Total Hours Worked: 45

Student's Signature: **Date:** 2082-10-16

Mentor's Signature: **Date:** 2082-10-16

Supervisor's Signature: _____ **Date:** _____

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Weekly Internship Logbook

Week No: 12

Name of Student: Darshan Dhakal

Internship Site/ Organization: Information Care Pvt. Ltd

Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-10-12	9	486	Fixed Outlook invitation email rendering issue.
2	2082-10-13	9	495	Implemented DKIM CSV download feature.
3	2082-10-14	9	504	Deployed updates to staging environment.
4	2082-10-15	9	513	Deployed updates to production environment.
5	2082-10-16	9	522	Final documentation and internship wrap-up.
6	2082-10-17	Holiday	522	Weekend
7	2082-10-18	Holiday	522	Weekend

Current Week Summary: Completed production improvements including Outlook email fix and DKIM CSV feature deployment. Finalized documentation and handover.

Next Week Plan: Internship completed.

Total Hours Worked: 45

Student's Signature: **Date:** 2082-10-16

Mentor's Signature: **Date:** 2082-10-16

Supervisor's Signature: _____ **Date:** _____